

Date: 12/09/2025

Lab Practical #14:

Implementation of parity bit check Using C/Java language with example.

Practical Assignment #14:

1. C/Java Program: Implementation of parity bit check Using C/Java language.

```
import java.util.Scanner;

public class ParityBitCheck {

    public static String addParityBit(String dataBits, String parityType) {
        int countOnes = 0;
        for (char bit : dataBits.toCharArray()) {
            if (bit == '1') countOnes++;
        }

        char parityBit;

        if (parityType.equalsIgnoreCase("even")) {
            parityBit = (countOnes % 2 == 0) ? '0' : '1';
        } else {
            parityBit = (countOnes % 2 == 0) ? '1' : '0';
        }

        return dataBits + parityBit;
    }

    public static boolean checkParity(String data, String parityType) {
        char parityBit = data.charAt(data.length() - 1);
        String dataBits = data.substring(0, data.length() - 1);

        int countOnes = 0;
        for (char bit : dataBits.toCharArray()) {
            if (bit == '1') countOnes++;
        }

        boolean isValid;
        if (parityType.equalsIgnoreCase("even")) {
```

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```
        isValid = ((countOnes % 2 == 0) && parityBit == '0') ||  
                ((countOnes % 2 == 1) && parityBit == '1');  
    } else {  
        isValid = ((countOnes % 2 == 0) && parityBit == '1') ||  
                ((countOnes % 2 == 1) && parityBit == '0');  
    }  
  
    return isValid;  
}  
  
public static void main(String[] args) {  
    Scanner scanner = new Scanner(System.in);  
  
    System.out.print("Enter binary data (without parity bit): ");  
    String dataBits = scanner.nextLine();  
  
    System.out.print("Enter parity type (even/odd): ");  
    String parityType = scanner.nextLine();  
  
    String dataWithParity = addParityBit(dataBits, parityType);  
    System.out.println("Data with " + parityType + " parity bit: " +  
dataWithParity);  
  
    if (checkParity(dataWithParity, parityType)) {  
        System.out.println("Parity check PASSED.");  
    } else {  
        System.out.println("Parity check FAILED.");  
    }  
  
    scanner.close();  
}  
}
```